

Arvin Munkh-Ochir

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Skills

Programming Languages : Embedded Rust, React.js, LuatOS, HTML, CSS, JavaScript, JAVA, Python, C++

Engineering Skills: Transmission Protocols: UART, SPI, I²C

Software & Tools: OpenBCI, OpenVibe, CorelDRAW, InkScape

Experience

Software Engineering Intern, OndoSpace LLC – Ulaanbaatar, Mongolia June 2025 – Jul 2025

- Researched AI applications at the intersection of space and telecom at ONDOSpace, initiating the ongoing GerGrid AI project.
- Launched SugarBox, an internal feedback platform built with React and Next.js, adopted by 80% of ONDO LLC employees.

Electrical and Computer Engineering Intern, OndoSpace LLC – Ulaanbaatar, Mongolia Jun 2024 – Dec 2024

- Improved motion sensing accuracy for 0.5 U Cube Satellite Development to receive accurate calibrated data from 9-axis IMU Sensor, by programing them on Embedded Rust.
- Contributed to Mongolian Pioneering Satellite Mission Development through designing mission patches and “Remove Before Flight” tags, by drawing them on CorelDraw.
- Evaluated the data usability for Satellite development executing test comparison codes on BMX160, HW038, MCP9808 STM32 processors and sensors, by coding them on Embedded Rust.

Full Time SAT Mathematics teacher, TOMUJIN Academy LLC – Ulaanbaatar, Mongolia Sept 2023 – May 2024

- Led four SAT Math programs as measured by teaching 200+ students in total, through rural areas of Mongolia. Boosted student SAT Math scores with result of average improvements of 380 points.

Projects

GerGrid AI Infrastructure Planning Start-Up

- Explored the usage of AI based models to optimize cell tower placement in Mongolia’s mountainous terrain by integrating existing LLM models with telecommunication platforms like Atol. Included ONDOSpace’s LEO orbited cube satellites feature.

Spelling Letters Through Brain Waves

[Arvin/projectbci](#)

- Explored brain computers interface with non-invasive EEG technology to detect letters from brain signals. Implementing the P300 speller paradigm of OpenVibe Software, achieved accuracy improvement from 72% to 82% through repeated training.

Location Detecting Smart Backpack

[Arvin/projectcane](#)

- Developed a smart backpack for visually impaired users with integrated location sensing, time announcements, and location sharing. Designed to enhance safety, independence, and daily navigation in dynamic urban environments, through programming sensors with LuatOS.

Smart Cane for Visually Impaired People

[Arvin/projectbackpack](#)

- Built an early smart cane prototype for visually impaired users, integrated five infrared sensors and a water sensor with a microcontroller. Enabled real time audio feedback to detect water, obstacles, and surface holes, by using embedded system’s async features.

Hamtdaa - Content Creation

[Arvin/Hamtdaa](#)

- Self driven initiative to improve rural areas education by creating content that combines international level education with Mongolian language. Resulted in reaching 1500 people as in overlapped numbers

Education

School – Stony Brook University | BS in Computer Science June 2029

School – Orchlon International High School June 2023